

Program – 7: Deploy a Web Application to Kubernetes using Minikube

Description

This experiment demonstrates deploying an Nginx web server on a Kubernetes cluster using a Deployment and exposing it using a Service in Minikube.

Prerequisites

- Minikube installed
- kubectl installed

Procedure

Step 1: Start Minikube

minikube start

Step 2: Check Minikube Status

minikube status

Step 3: Create Deployment (nginx-deployment.yaml)

apiVersion: apps/v1

kind: Deployment

metadata:

name: nginx-deployment

spec:

replicas: 2

selector:

matchLabels:

app: nginx

template:

metadata:

labels:

app: nginx

spec:

containers:

- name: nginx

image: nginx:latest

ports:

- containerPort: 80

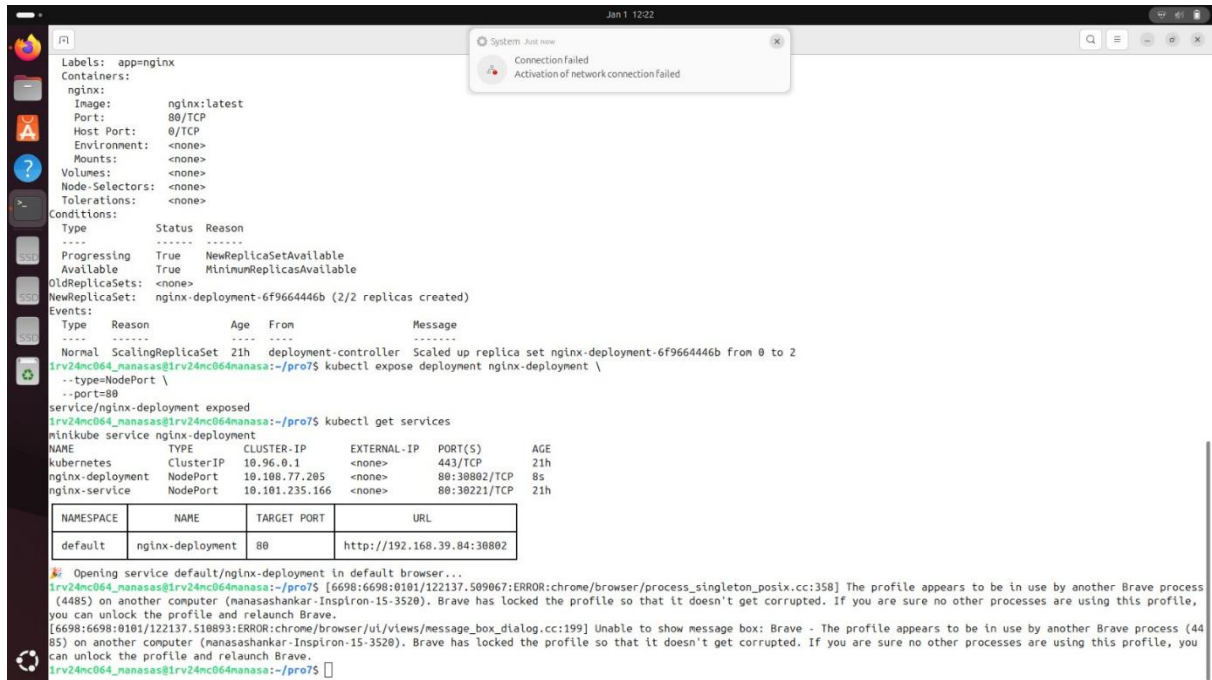
Step 4: Apply Deployment

kubectl apply -f nginx-deployment.yaml

Step 5: Verify Deployment and Pods

kubectl get deployments

kubectl get pods



```
Labels: app=nginx
Containers:
  nginx:
    Image:      nginx:latest
    Port:      80/TCP
    Host Port:  80/TCP
    Environment: <none>
    Mounts:      <none>
    Volumes:     <none>
    Node-Selectors: <none>
    Tolerations: <none>
Conditions:
  Type           Status  Reason
  ----           -
  Progressing    True    NewReplicaSetAvailable
  Available      True    MinimumReplicasAvailable
  OldReplicaSets: <none>
  NewReplicaSet:  nginx-deployment-6f9664446b (2/2 replicas created)
Events:
  Type    Reason              Age    From                      Message
  ----    -
  Normal  ScalingReplicaSet   21h    deployment-controller     Scaled up replica set nginx-deployment-6f9664446b from 0 to 2
irv24mc064_nanasas@irv24mc064manasa:~/pro7$ kubectl expose deployment nginx-deployment \
--type=NodePort \
--port=80
service/nginx-deployment exposed
irv24mc064_nanasas@irv24mc064manasa:~/pro7$ kubectl get services
NAME                                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
kubernetes                          ClusterIP   10.96.0.1      <none>          443/TCP    21h
nginx-deployment                    NodePort    10.108.77.205  <none>          80:38882/TCP  8s
nginx-service                       NodePort    10.101.235.166 <none>          80:38221/TCP  21h
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.39.84:38882

```
Opening service default/nginx-deployment in default browser...
irv24mc064_nanasas@irv24mc064manasa:~/pro7$ [6698:6698:0101/122137.509067:ERROR:chrome/browser/process_singleton_posix.cc:358] The profile appears to be in use by another Brave process
(4485) on another computer (manasashankar-Inspiron-15-3520). Brave has locked the profile so that it doesn't get corrupted. If you are sure no other processes are using this profile,
you can unlock the profile and relaunch Brave.
[6698:6698:0101/122137.510893:ERROR:chrome/browser/ui/views/message_box_dialog.cc:199] Unable to show message box: Brave . The profile appears to be in use by another Brave process (44
85) on another computer (manasashankar-Inspiron-15-3520). Brave has locked the profile so that it doesn't get corrupted. If you are sure no other processes are using this profile, you
can unlock the profile and relaunch Brave.
irv24mc064_nanasas@irv24mc064manasa:~/pro7$
```

Step 6: Expose Deployment as a Service

kubectl expose deployment nginx-deployment --type=NodePort --port=80

Step 7: Access the Web Application

minikube service nginx-deployment

This opens the Nginx web page in your browser.

Step 8: Clean Up Resources

kubectl delete pods --all

kubectl delete svc --all

kubectl delete deployments --all

Expected Output

Nginx default welcome page displayed in the browser.



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Result

Successfully deployed and accessed a web application on Kubernetes using Minikube.